Report on

“Tutorials”

For

Software Project Management - 4CS504ME25

B. Tech. Semester IV

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Aim

To study and compare the major Software Development Process Models - Waterfall, Incremental, Spiral, Agile, and DevOps - and to justify the most suitable process model for the Next-Generation Hospital Management System (HMS) project from a Software Project Management perspective.

Objective

- Understand the working principles of the Waterfall, Incremental, Spiral, Agile, and DevOps models.
- Compare the models on the basis of flexibility, risk handling, cost, and suitability.
- Identify the advantages and disadvantages of each model.
- Map the characteristics of the HMS project (scope, constraints, team size, timeline) to the most appropriate process model.
- Justify the selection of the chosen model with supporting reasoning.

1. Introduction

A software development process model defines the sequence of activities involved in software creation, from requirement gathering to deployment and maintenance. Selecting an appropriate process model is one of the most important early decisions in software project management, as it directly affects project cost, schedule, quality, and the team's ability to respond to change.

For the Next-Generation Hospital Management System (HMS) project - which involves multiple interacting modules such as patient registration, appointment scheduling, pharmacy stock, and billing, developed by a small team under a fixed academic timeline - the choice of process model needs to balance structure with flexibility. This report compares five widely used models and recommends the one best suited to the HMS project.

2. Overview of Process Models

2.1 Waterfall Model

A sequential model where each phase is completed before the next begins, best suited for clear and stable requirements.

2.2 Incremental Model

A model that develops the system in small increments, with each build adding and delivering new functionality.

2.3 Spiral Model

A risk-driven model that develops the system in cycles, with each cycle involving planning, risk analysis, development, and evaluation.

2.4 Agile Model

An iterative approach that delivers working software in short sprints, focusing on feedback, collaboration, and adapting to change.

2.5 DevOps Model

An approach that combines development and operations through CI/CD automation, enabling continuous integration, delivery, and collaboration.

Criteria	Waterfall	Incremental	Spiral	Agile	DevOps
Approach	Linear & sequential	Sequential builds, each adding features	Iterative + risk-driven cycles	Iterative, sprint-based	Continuous, automated pipeline
Flexibility to change	Very low	Moderate	Moderate to high	Very high	High
Risk handling	Poor - risks found late	Better than Waterfall	Excellent - dedicated risk analysis	Good - reviewed every sprint	Good - issues caught via automation
Customer involvement	Only at start & end	After each increment	At each spiral review	Continuous	Continuous
Documentation	Heavy	Moderate	Heavy	Minimal, just enough	Automated / lightweight
Cost of change	Very high (late stages)	Moderate	Moderate (managed via risk analysis)	Low	Low
Delivery speed	Slow (single release)	Moderate (phased releases)	Slow (long spiral cycles)	Fast (frequent releases)	Very fast (continuous releases)
Best suited for	Small, well-defined projects	Projects with evolving but known modules	Large, complex, high-risk projects	Projects needing frequent feedback	Projects needing rapid, reliable releases
Team size	Any	Small to medium	Large, experienced	Small to medium, cross-functional	Small to medium, cross-functional

3. Comparison of Process Models

4. Advantages and Disadvantages

4.1 Waterfall

Advantages:

- Simple to understand and manage.
- Clear milestones and documentation at each stage.

Disadvantages:

- Very difficult to accommodate changing requirements.
- Working software is available only at the very end.

4.2 Incremental

Advantages:

- Working modules delivered early and regularly.
- Easier to test and debug smaller increments.

Disadvantages:

- Requires good upfront planning to define increments.
- Total cost can exceed Waterfall if not planned well.

4.3 Spiral

Advantages:

- Strong emphasis on risk identification and mitigation.
- Suitable for large, evolving, high-risk systems.

Disadvantages:

- Complex to manage and requires risk-analysis expertise.
- Can be costly and time-consuming for small projects.

4.4 Agile

Advantages:

- Adapts quickly to changing requirements.
- Frequent delivery of working software builds confidence.
- Continuous stakeholder feedback improves quality.

Disadvantages:

- Requires active customer/stakeholder involvement throughout.
- Less emphasis on documentation can affect long-term maintenance.

4.5 DevOps

Advantages:

- Fast, automated, and reliable releases.
- Close dev-ops collaboration reduces deployment issues.

Disadvantages:

- Requires investment in automation tooling and CI/CD setup.
- Needs cultural and process maturity across teams.

5. Justification of the Most Suitable Model for the HMS Project

The HMS project is best suited for an **Agile (Scrum) process model with incremental delivery**. Since the system is divided into six different modules, the team can develop and test each module step by step in different sprints. Agile also gives flexibility to make changes based on mentor and faculty feedback during the project.

It is also suitable for our small two-member team and fixed semester time. Regular sprint reviews helps us to find problems early, especially in areas like multi-tenant architecture. DevOps practices like regular testing and builds can be used, but a complete DevOps model is not required for this academic project.

Conclusion:

Agile is selected because it supports sprint-wise development, easy changes, early feedback, and suits our small team and fixed academic timeline.